

Multiple Choice (10 marks)

1. Let $\{N(t), t \in [0, \infty)\}$ be a Poisson process with rate of $\lambda = 4$ and X_1 be the first arrival time. Given $N(t) = 1$, then what is $P(X_1 \leq t/2)$?
 - (a) 0.5
 - (b) 1
 - (c) 0.95
 - (d) 0.65
 - (e) 0.85
2. Consider the initial value problem $4y'' + 12y' + 9y = 0$, $y(0) = 1$, $y'(0) = -4$. Determine where the solution has the value zero.
 - (a) 0.8
 - (b) 0.3
 - (c) 0.1
 - (d) 0.5
 - (e) 0.4
3. Statement 1 | A factor group of a non-Abelian group is non-Abelian. Statement 2 | If K is a normal subgroup of H and H is a normal subgroup of G , then K is a normal subgroup of G .
 - (a) False, False
 - (b) True, True
 - (c) Not Determinable, True
 - (d) Not Determinable, False
 - (e) True, False
4. Sam paid \$8.28 for 18 stamps. At this rate, how much would it cost Sam to buy 12 stamps?
 - (a) \$4.52
 - (b) \$7.28
 - (c) \$5.52
 - (d) \$6.28
 - (e) \$3.72

5. The graph of the equation $x + 2y + 3 = 0$ is perpendicular to the graph of the equation $ax + 2y + 3 = 0$. What is the value of a ?
- (a) -1
 - (b) -13
 - (c) -4
 - (d) -3
 - (e) 0

True / False (10 marks)

Answer True or False

- 6. True or False: The limit of the sequence a_n as n approaches infinity is 0.5, given the conditions $0 < a_n < 1$ and $(1-a_n)a_{n+1} > 1/4$.
- 7. True or False: The value of the expression $-25 - (-11)$ is -11, indicating a misunderstanding of the operation's effect on negative values.
- 8. True or False: Every nonzero free abelian group indeed has an infinite number of bases due to the independence of its elements.
- 9. True or False: The determinant of matrix $A = \begin{bmatrix} 2 & 4 & 3 \\ 3 & 3 & 1 \\ 42 & 20 & 51 \end{bmatrix}$ is 200.
- 10. True or False: The difference of 142.76 and 16.5 is accurately calculated as 126.26.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. The points $A(5, -5)$ and $B(-1, -1)$ are the endpoints of the hypotenuse of an isosceles right triangle $\triangle ABC$. What is the area of ABC ?
- 12. Analyze the set $\{\frac{2}{5}, \frac{1}{2}, 5, 10\}$ and determine the maximum possible value of the quotient $\frac{x}{y}$ for any two distinct elements. Justify your reasoning.

End of Paper